- *Lightweight construction*: Early drones were constructed from wood, fabric, and early aluminum alloys, materials chosen for their light weight and ease of fabrication. This lightweight design made them easier to launch and maneuver, but also limited their structural integrity and payload capacity. The *Queen Bee (1935)*, for instance, utilized a wooden fuselage covered with fabric to reduce cost and weight, setting a standard for early UAV design simplicity.
- *Mechanical simplicity*: The mechanical systems of these drones were extremely basic, consisting of servo-driven linkages and analog control mechanisms. They lacked avionics, processors, or any automated components. As a result, pilots had to compensate manually for turbulence, wind, and other aerodynamic disturbances, making the operation highly challenging. ••
- *Absence of sensors or autonomy*: First-generation drones were devoid of any sensing or autonomous capabilities. They operated without gyroscopes, altimeters, or navigational aids such as GPS. Navigation depended entirely on the operator's visual observation and manual corrections during flight, often making long or complex missions impractical. ••

Figure 2.1 illustrates the fundamental design and operational characteristics of first-generation drones, showcasing their simplicity and manual mode of operation. These early UAVs were primarily fixed-wing aircraft constructed from lightweight materials such as wood, canvas, or early aluminum alloys, emphasizing structural efficiency over endurance. The figure highlights key features, including manual control, where operators relied entirely on basic radio transmitters to steer and stabilize the drone in flight. The limited range and endurance of these systems were